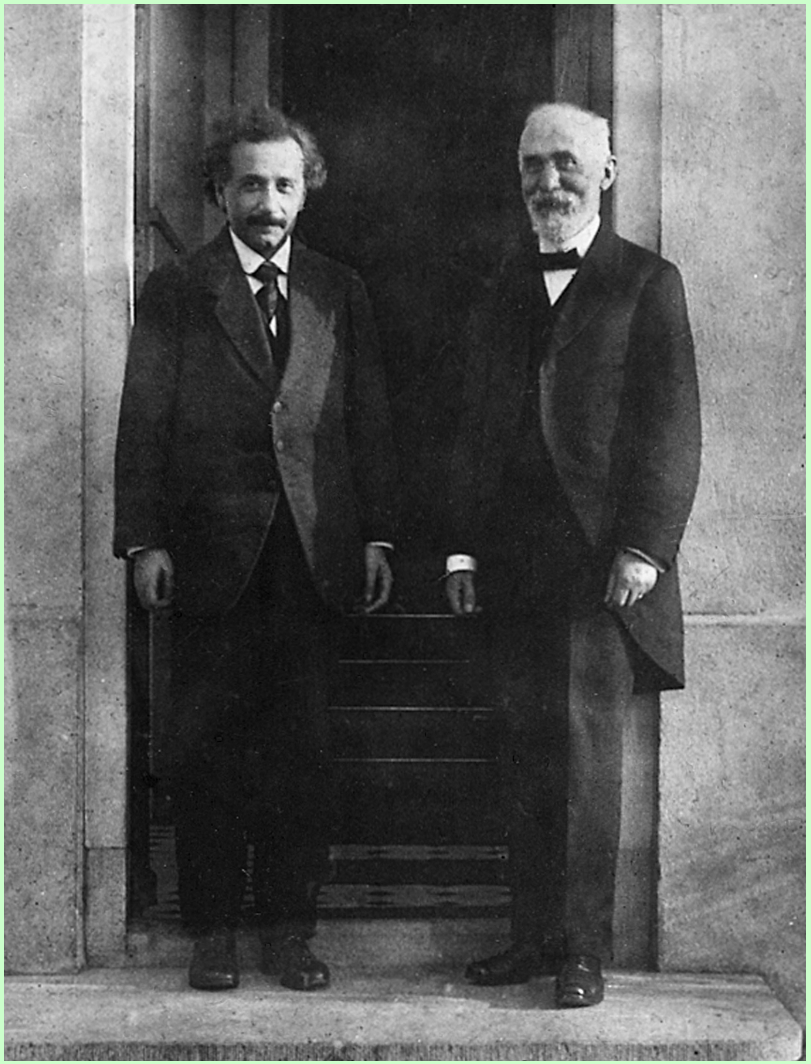
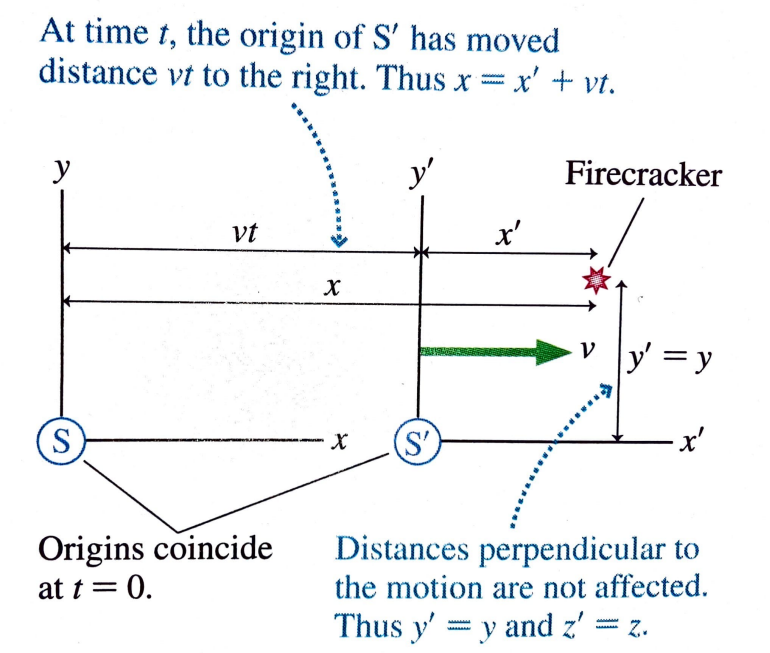
Albert Einstein



Hendrik Lorentz

Lorentz Transformation

Recall: Galilean Relativity



$$
\begin{aligned}
t' &= t \\
x' &= x - vt \\
y' &= y \\
z' &= z
\end{aligned}
$$

Full set of transformations

We want a similar relativistic transformation. Lorentz figured it out

$$\begin{aligned} t' &= \gamma\left(t - \frac{vx}{c^2}\right) \\ x' &= \gamma\left(x - vt\right) \\ y' &= y \\ z' &= z \end{aligned}$$

The Lorentz Transformation
You have everything in S, but
Want to know about S' which
Is moving +x from S perspective

$$\begin{aligned} t &= \gamma\left(t' + \frac{vx'}{c^2}\right) \\ x &= \gamma\left(x' + vt'\right) \\ y &= y' \\ z &= z' \end{aligned}$$

The Inverse Lorentz Transformation
You have everything in S', but
Want to know about S which
Is moving -x from S' perspective

Note you can just replace v with -v

Ex. Show that the Lorentz transformation reduces to the **time dilation equation** for two measurements done with the same x'
Show that the Lorentz transformation reduces to the **length contraction equation** for two measurements done with the same t'

The Lorentz Transformation has a lot going on! But we can simplify greatly
If we properly scale our t and x axes by using comparable units. So
If you just use units of (t, x) = (seconds, lightseconds) or years and lightyears…

$$\begin{aligned} t' &= \gamma(t - \beta x) \\ x' &= \gamma(x - \beta t) \\ y' &= y \\ z' &= z \end{aligned}$$

Properly scaled Lorentz Transformation

$$\begin{aligned} t &= \gamma\left(t' + \beta x'\right) \\ x &= \gamma\left(x' + \beta t'\right) \\ y &= y' \\ z &= z' \end{aligned}$$

Inverse Transformation
(just replace β with - β)

Even simpler if we rewrite as matrix equations

Lorentz Transformation

$$\begin{aligned} t^{'} &= \gamma\left(t-\beta x\right) \\ x^{'} &= \gamma\left(x-\beta t\right) \\ y^{'} &= y \\ z^{'} &= z \end{aligned}$$

$$\begin{pmatrix} t^{'} \\ x^{'} \\ y^{'} \\ z^{'} \end{pmatrix} = \begin{pmatrix} \gamma & -\gamma\beta & 0 & 0 \\ -\gamma\beta & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} t \\ x \\ y \\ z \end{pmatrix}$$

$$x^{'} = \Lambda x$$

Inverse Lorentz Transformation

$$\begin{aligned} t &= \gamma\left(t^{'}+\beta x^{'}\right) \\ x &= \gamma\left(x^{'}+\beta t^{'}\right) \\ y &= y^{'} \\ z &= z^{'} \end{aligned}$$

$$\begin{pmatrix} t^{'} \\ x^{'} \\ y^{'} \\ z^{'} \end{pmatrix} = \begin{pmatrix} \gamma & \gamma\beta & 0 & 0 \\ \gamma\beta & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} t \\ x \\ y \\ z \end{pmatrix}$$

$$x = \Lambda^{-1} x^{'}$$

Even better, we normally don't care about y and z, only x and t so we can use The 2d version

Lorentz Transformation

$$\begin{pmatrix} t' \\ x' \end{pmatrix} = \begin{pmatrix} \gamma & -\gamma\beta \\ -\gamma\beta & \gamma \end{pmatrix} \begin{pmatrix} t \\ x \end{pmatrix}$$

$$\boxed{x' = \Lambda x}$$

Inverse Lorentz Transformation

$$\begin{pmatrix} t \\ x \end{pmatrix} = \begin{pmatrix} \gamma & \gamma\beta \\ \gamma\beta & \gamma \end{pmatrix} \begin{pmatrix} t' \\ x' \end{pmatrix}$$

$$\boxed{x = \Lambda^{-1} x'}$$

Ex. Abby is on a train car with firecrackers at both ends. The traincar moves past Ben outside with v = 0.8c. Both flashes reach Ben simultaneously 1 μs after Abby passes. He later finds burn marks from the firecracker going off 300 m on either side of him. Determine the spacetime coordinates for the Left and Right firecracker events.

Ex. An observer on a rocket moving in the +x direction past Earth at v = 0.6c sees a collision between two comets at (200s, 3×10^{10} m) in her Frame. Call this event C. What are the spacetime coordinates of the collision in the Earth frame?

Ex. Event A happens at (t,x) =(3,5). What are the spacetime coordinates of the event in a frame that is moving with v = +0.9c with respect to the first frame

An example where we solve the same problem with ST diagrams AND Lorentz Transformation